

The European single market and intra-EU trade: an assessment with heterogeneity-robust difference-in-differences methods

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Abstract

We use heterogeneity-robust difference-in-differences (DiD) methods to evaluate the impact of membership in the European Union (EU) single market on international trade. On the policy front, we provide evidence that: (i) on average, the EU has been very effective in promoting trade among its member states; (ii) the trade effects of the EU have been long-lasting, but heterogeneous across EU cohorts; and (iii) while the EU has benefited both ‘old’ and ‘new’ members, the increase in the exports from the ‘old’ members to the ‘new’ joiners has been disproportionately larger. From a methods and practical perspective, the contribution of this paper is to introduce a new, fast and flexible estimation command that combines leading estimation techniques from the gravity literature with recent methods from the heterogeneity-robust DiD literature.

KEYWORDS

EU membership, staggered difference-in-differences, gravity model, estimation command

JEL CLASSIFICATION

C13; C23; F10; F13; F14

1 | INTRODUCTION

On 1 May 2024, the European Union (EU) celebrated the 20th anniversary of its largest enlargement in terms of number of states and population.¹ On the eve of this anniversary, the EU is considered the most prominent and successful international integration effort in the world: ‘It is also the world’s largest single market area, and free trade among its members was one of the EU’s founding principles’.² Thus, perhaps not surprisingly, the most important trade partners of most EU members are other EU states.

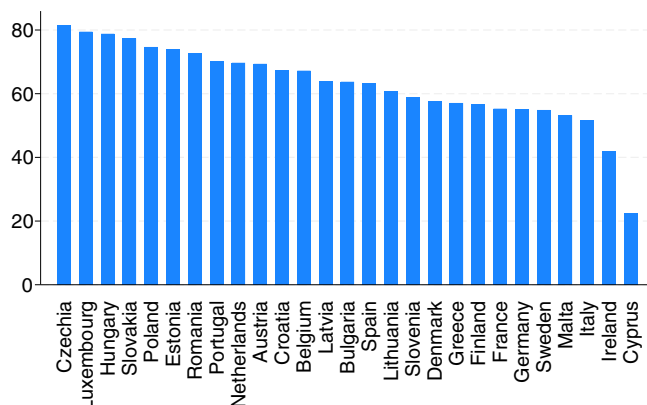


FIGURE 1 Intra-EU exports of goods in 2023 (% of total). *Notes:* The figure reports the share of intra-EU exports of goods in 2023 as a percentage of total exports. The data come from Eurostat (2023).

Figure 1 captures two notable patterns of intra-EU trade of goods in 2023. First, more than 50% of the exports for 25 of the 27 EU members are directed to other EU members. The exceptions are Cyprus and Ireland. Second, the country-specific shares of intra-EU trade are very heterogeneous, ranging from 22% and 42% for Cyprus and Ireland, respectively, to 81% and 79% for Czechia and Luxembourg.

While few would doubt the significant role of the EU single market to promote trade among its members, the following two, more nuanced, questions should be of interest to policymakers and academics alike. What fraction of the large shares of intra-EU trade is actually due to joining the single market as opposed to other forces, such as geographic and cultural proximity? Has the single market succeeded in promoting intra-EU trade uniformly, or did some members enjoy larger gains from free trade within the EU? Answering these questions is important for *ex post* evaluation of the success of EU trade integration, but also for *ex ante* evaluation of the potential implications of further enlargement of the EU. Moreover, obtaining reliable partial estimates of the effects of the single market on trade is a crucial prerequisite for estimating sound general equilibrium and welfare implications of joining or leaving the EU (Dhingra *et al.* 2017).

Despite the economic and political importance of the EU as the largest free trade area in the world, there are surprisingly few studies that have evaluated the impact of EU membership on intra-EU trade (e.g. Hamilton and Winters 1992; Fontagné *et al.* 1998; Head and Mayer 2000; Papazoglou *et al.* 2006; Iliev *et al.* 2016). Moreover, all studies of the impact of the EU that we reviewed relied exclusively on traditional ‘gravity-type’ methods,³ and we are not aware of any investigation that accounts for possible biases in the presence of treatment effect heterogeneity, for example, due to so-called ‘forbidden comparisons’ that (mis)use already-treated units in the control group (e.g. Borusyak and Jaravel 2017; de Chaisemartin and D’Haultfœuille 2020).

This caveat may be particularly important for the evaluation of the effects of EU membership, which is a prominent example of a staggered policy adoption design, where new members have joined in waves of accession that, as depicted in Figure 2, also vary in the number of new members—for example, the large 2004 wave versus the accession of Bulgaria and Romania in 2007 versus the accession of Greece in 1981 or Croatia in 2013.⁴

Against this backdrop, the contribution of our paper is twofold. First, from a policy and academic perspective, we combine established methods from the structural gravity literature with recent heterogeneity-robust difference-in-differences (DiD) methods to re-evaluate the impact of membership in the EU single market on international trade, and to explore the heterogeneity of the EU effects across various dimensions. Second, on the methods and practical front, our contribution is to introduce a new, fast and flexible estimation command

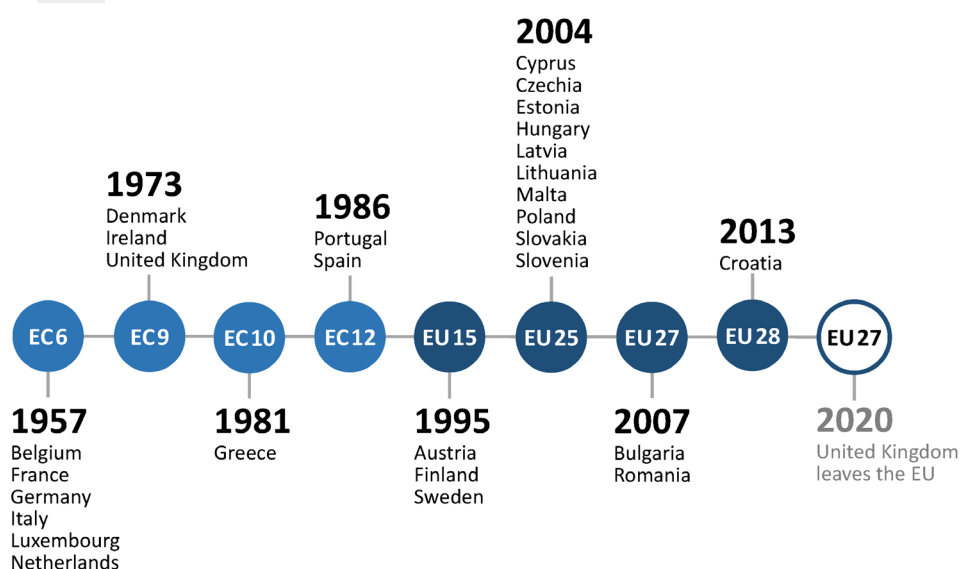


FIGURE 2 EU accession timeline. *Notes:* The figure traces the accession timeline for new EU members. EC denotes the European Community, which was the predecessor of the EU. The layout and style of the figure is based on the corresponding figure from the German Federal Statistical Office at https://www.destatis.de/Europa/EN/Country/EU-Member-States/_EU_EZ_Zeitverlauf_en.html (accessed 23 October 2025).

that combines leading estimation techniques from the gravity literature with recent methods from the heterogeneity-robust DiD literature and fast computation approaches to handle high-dimensional fixed effects in linear and non-linear econometric models.

To build our command, we capitalize on developments from three influential strands of the literature. First, we take into account many recent contributions from the vibrant DiD econometrics literature (e.g. Hull 2018; de Chaisemartin and D'Haultfœuille 2020, 2023; Callaway and Sant'Anna 2021; Goodman-Bacon 2021; Sun and Abraham 2021; Wooldridge 2021, 2023; Borusyak *et al.* 2024; Goldsmith-Pinkham *et al.* 2024). Second, we take advantage of the recent developments for fast computation and convergence of linear and non-linear models with high-dimensional fixed effects (e.g. Correia, 2016b; Correia *et al.* 2020). Third, we rely on the empirical and theoretical trade gravity literature (e.g. Eaton and Kortum 2002; Anderson and van Wincoop 2003; Arkolakis *et al.* 2012; Yotov *et al.* 2016; Nagengast and Yotov 2025) for guidance and recommendations for some of the main options of the command.

The result is a fast and flexible estimation command for heterogeneity-robust DiD estimations—`jwddid`.⁵ The command implements the extended two-way fixed effects (ETWFE) estimator proposed by Wooldridge (2021, 2023), which is robust to treatment effect heterogeneity in settings of staggered DiD by introducing suitable cohort-year treatment effects. In the presence of heterogeneous treatment effects, the standard two-way fixed effects (TWFE) estimator can yield biased and hard-to-interpret estimates because it aggregates cohort-year treatment effects with implicit weights that may in extreme cases even be negative (Borusyak *et al.* 2024; de Chaisemartin and D'Haultfœuille 2020). Our current analysis focuses on the effects of EU membership on international trade. However, the proposed command is much more general. Naturally, it can be applied to quantify the impact of other determinants of trade flows within the same trade gravity setting (e.g. World Trade Organization (WTO) membership, currency unions). Moreover, it can be used to estimate gravity equations for other bilateral flows and relationships (e.g. migration, foreign direct investment (FDI), patents, international M&A). Finally, the command can also be used in more general (non-gravity) panel and cross-sectional settings (e.g. Liebersohn 2024; Ahmadi *et al.* 2024).

To perform the empirical analysis, we rely on several standard trade data sources. First, we secure a maximum number of observations over a long period of time by using a combination of the *Direction of Trade Statistics* (DoTS) of the International Monetary Fund (IMF) and the United Nations *Commodity Trade Statistics* (COMTRADE) databases. The resulting estimating sample covers aggregate trade flows for 260 countries over the period 1950–2019.⁶ The long time coverage of these data enables us to obtain estimates of the EU effects for recent members as well as the founding countries.

In addition to the aggregate trade data, we also employ the *Structural Gravity Database* (SGD) of the WTO, which covers manufacturing trade for 229 countries over the period 1980–2016. An advantage of the SGD sample is that it complements our aggregate estimates with sectoral results for manufacturing, which is by far the most traded sector within the EU.⁷ Another advantage of the SGD is that it includes consistently constructed domestic trade flows, which may be important for quantifying the impact of trade policies (Yotov 2022).

We draw several conclusions based on the empirical analysis. First, we find that membership in the EU has led to a significant increase in bilateral trade flows among its member states. Specifically, our preferred estimate implies that membership in the EU has led to an increase of 52% in the bilateral trade flows between its member states. From a methodological perspective, our preferred ETWFE estimate is significantly larger than the corresponding TWFE result that is obtained with traditional gravity methods. We also note that both the ETWFE and TWFE EU estimates that we obtain are larger than recent estimates of the effects of trade agreements, thus reinforcing the perception of the EU as one of the most successful free trade areas in the world. Finally, an event-type analysis reveals that the effects of the EU on members' trade are also significantly more long-lasting as compared to the corresponding effects of trade agreements from the related literature.

The average EU estimate that we obtain is masking significant heterogeneity across several dimensions. For example, we see significant differences in the EU effects on exports across cohorts. Specifically, with the exception of the 1981 cohort, the effects for the early cohorts (1957, 1973 and 1986) are larger than those for the late cohorts (1995, 2004, 2007 and 2013). We also observe heterogeneity in the evolution of the EU effects across cohorts. For example, the effects for the 1973 and 1986 cohorts are large and are exhausted relatively quickly, while the effects for the 1995 cohort are smaller, but more long-lasting. Finally, we find that the effects of the EU have been disproportionately stronger for the exports from 'old' members to 'new' members than in the opposite direction, while the effects between 'new' members have been almost as strong as those between 'old' and 'new' members.

Our overall conclusion is that the effects of the EU have been very strong on average, and that the EU has benefited all members; however, the EU effects have been very heterogeneous. In addition, our asymmetric estimates point to possible areas for policy intervention.

We also test and confirm the robustness of our results and main conclusions to different datasets, and the degree of heterogeneity of the ETWFE model and alternative estimators, as well as DiD- and ETWFE-specific assumptions. First, we consider the WTO SGD as an alternative dataset, which includes domestic trade flows and which focuses on manufacturing trade, and we also perform several robustness checks related to the construction of our baseline dataset. Second, we consider restricting the heterogeneity of the ETWFE model, but we also use an imputation estimator that is more flexible. In addition, we consider a jackknife bias correction for potential incidental parameter problems, alternative standard error clusterings, and ordinary least squares (OLS) as an alternative estimator. Third, we consider the effect of methodological choices related to the definition of treatment onset, the potential effect of the presence of other treatments, the exclusion of long-term effects, a variant relaxing the parallel trends assumption, and alternative weighting schemes for obtaining the aggregate treatment effect coefficient. Although the size of the coefficient varies slightly in some cases, all robustness tests leave the estimates of EU membership on trade by and large unchanged in terms of sign and significance.

The rest of the paper is organized as follows. Section 2 discusses the current methods to estimate trade gravity models, and draws implications and recommendations for the key features of our estimation command. Section 3 introduces the new command by describing its key building blocks and main features. Section 4 offers a brief description of our data, the data sources and some key summary statistics. Section 5 presents our main findings and the results from a series of validity checks and sensitivity experiments. Section 6 describes the results from a series of robustness experiments. Section 7 concludes. Finally, an Online Appendix includes a detailed description of the new command, including all of its options and some additional estimation results.

2 | ESTIMATING THE EFFECTS OF THE EU ON TRADE

The objective of this section is to review the current and established techniques and developments from the empirical trade literature for estimating the effects of the EU (and other policies) on international trade, and to draw implications for the new estimation command, which we introduce in the next section. To this end, we rely on the current recommendations from the structural gravity and the staggered DiD literature, as recently summarized by Nagengast and Yotov (2025), and we specify the following *structural*⁸ estimating gravity equation that is used to obtain our estimate of the effects of EU membership on international trade:

$$X_{ij,t} = \exp \left[\phi_{i,t} + \psi_{j,t} + \vec{\gamma}_{ij} + \sum_{g=q}^T \sum_{s=g}^T \delta_{gs} D_{gs} \right] + \varepsilon_{ij,t}, \quad \text{for all } i, j, t. \quad (1)$$

The dependent variable in equation (1), $X_{ij,t}$, denotes nominal exports (at delivered prices) from exporter i to importer j at time t .⁹ Consistent with the staggered DiD methods, and due to availability of better data, most of the recent gravity applications are performed in panel settings. In addition, we follow the recommendations of Egger *et al.* (2022) to estimate our model with data for consecutive years. Historically, in response to the critique of Cheng and Wall (2005, p. 52, fn. 8) that ‘[f]ixed-effects estimation is sometimes criticized when applied to data pooled over consecutive years on the grounds that dependent and independent variables cannot fully adjust in a single year’s time’, many gravity applications (e.g. Baier and Bergstrand 2007; Olivero and Yotov 2012) have used interval data instead of data for consecutive years. However, as discussed in Egger *et al.* (2022), the use of intervals instead of consecutive years requires dropping data randomly, and may prevent estimation of the effects of some policies, especially when the focus is on their heterogeneous effects, which often are identified from relatively few observations. The implication of using consecutive-year data for our new command is that it should be able to handle ‘long’ panel data and the corresponding fixed effects.

Due to the influential work of Santos Silva and Tenreyro (2006), the Poisson pseudo maximum likelihood (PPML) estimator has established itself as the leading gravity estimator. As originally motivated by Santos Silva and Tenreyro (2006), the two main advantages of the PPML estimator for gravity estimations are that (i) it can handle heteroscedasticity in the trade data, and (ii) due to its multiplicative form, it can handle zero trade flows. More recently, several papers document additional favourable properties of the PPML estimator for gravity estimations, which have reinforced its use as the leading estimator for gravity estimations. An important implication for our command is that it has to allow for PPML, while also providing the option for the estimation to be performed by OLS (and other estimators) for robustness.

Turning to the covariates in our model, equation (1) includes three sets of fixed effects. The theoretical motivation behind the exporter-time and importer-time fixed effects ($\phi_{i,t}$ and $\psi_{j,t}$, respectively) is that they control for the structural multilateral resistance terms of Anderson and van Wincoop (2003), whose omission has been labelled ‘the gold medal mistake’ in gravity

regressions by Baldwin and Taglioni (2006).¹⁰ In addition to the multilateral resistance terms, and from a broader econometric perspective, the exporter-time and importer-time fixed effects account for any other country-specific characteristic, such as country size, that may impact bilateral trade flows. The implication for our command is that gravity should be estimated with exporter-time and importer-time fixed effects.¹¹

The third set of fixed effects, $\vec{\gamma}_{ij}$, includes directional country-pair fixed effects. This follows directly from the DiD specification under consideration; however, there are also several appealing reasons to include those fixed effects in gravity regressions. First, as demonstrated by Egger and Nigai (2015) and Agnosteva *et al.* (2019), the pair fixed effects match the bilateral trade costs in the gravity model much better than the ‘standard gravity variables’ (e.g. distance, contiguity). The latter do well in predicting relative trade costs, but they fail to capture the level of bilateral trade costs. Specifically, they underpredict the bilateral trade costs for poor countries, and overpredict them for more developed countries. Second, on a related note and following Wooldridge (2010), Baier and Bergstrand (2007) argue that the country-pair fixed effects control for and absorb most of the unobserved correlation between the potentially endogenous bilateral policy variables in the error term in gravity models, thus mitigating endogeneity concerns. Finally, in order to allow for asymmetric bilateral trade costs and asymmetric trade policy effects, Baier *et al.* (2019) argue that the pair fixed effects should be directional, that is, ij versus ji .

The use of asymmetric country-pair fixed effects has two implications for our command. First, the command should be able to handle a very large set of fixed effects; for example, with a balanced dataset for 100 countries, the number of relevant country-pair fixed effects is 9900.¹² Second, as demonstrated by Weidner and Zylkin (2021), PPML estimations with exporter-time, importer-time and country-pair fixed effects may suffer from incidental parameter problems (IPPs).

Weidner and Zylkin (2021) propose an adjustment based on Taylor expansions to correct for such biases. Relatedly, Jochmans (2017) uses a quasi-differencing approach, and Pfaffermayr (2021) implements jackknife and bootstrap confidence interval methods. Following Weidner and Zylkin (2021), Nagengast and Yotov (2025) implement a Monte Carlo simulation to study the IPP bias in a staggered DiD gravity setting, and they find that given the relatively large time dimension of their data, IPPs are not such a big issue for the coefficient estimates; however, the standard errors could be downward biased. To address potential IPP concerns, Nagengast and Yotov (2025) follow Dhaene and Jochmans (2015), Pfaffermayr (2021) and Weidner and Zylkin (2021) to implement a split-panel jackknife bias-correction procedure.

In general, equation (1) and the command also allow the inclusion of additional time-varying bilateral control variables. While this is a common practice in the gravity literature, from an econometric perspective, additional time-varying variables may be included only if they are unaffected by the treatment and are strictly exogenous. For example, in the presence of domestic trade flows, and following Bergstrand *et al.* (2015), the specification should also include a set of time-varying border variables, which take value 1 for international trade, and are equal to zero otherwise. These time-varying border variables would capture common (across countries) globalization trends, such as improvements in communication and transportation technologies. In principle, other treatments, such as tariffs, free trade agreements and economic sanctions, could also be included in the specification. However, in general, the current estimation framework is not designed to handle the additional complications introduced by the presence of multiple treatments or continuous treatments, so it is not advisable to include them.¹³ An alternative approach to testing the robustness of the results to the presence of other treatments that may potentially confound the main effect of interest is to omit conspicuous episodes from the analysis. We demonstrate this approach in Subsection 6.3.

Finally, and most important for our purposes, the key term $\sum_{g=q}^T \sum_{s=g}^T \delta_{gs} D_{gs}$, which captures the effects of EU membership on trade in our model, is motivated by the staggered DiD literature. Consistent with most existing studies, in our main analysis, we focus on the impact of the EU

on trade between member countries; that is, a country pair ij belongs to treatment cohort g if the condition that both countries were EU members was fulfilled for the first time in year g , q is the first year of the treatment of cohort g , T is the last year of the panel, D_{gs} is a time-varying treatment indicator equal to 1 for cohort g for $s = t$ in post-treatment years and 0 otherwise, and δ_{gs} captures the cohort-year specific treatment effects. For example, $D_{2004,2004}$ would be equal to 1 for the country pair Austria–Poland in the year 2004, since Poland joined the EU in 2004, and Austria had already been an EU member since 1995; $\delta_{2004,2004}$ would capture the EU effect on trade between all country pairs of the 2004 cohort in the first year; $\delta_{2004,2005}$ would capture the EU effect on trade between all country pairs of the 2004 cohort in the second year; and so on.

In addition to obtaining a single/common EU estimate, which is the most common practice in the existing literature, we also obtain a series of other EU estimates across various dimensions that have been of interest to scholars and policymakers. Specifically, we explore the heterogeneity of our estimates by event time, by cohort, and we also obtain directional estimates on trade between ‘old’ and ‘new’ EU members depending on the recency of their EU membership.

While there is no universally accepted best practice for clustering the standard errors in gravity models, Egger and Tarlea (2015) and Pfaffermayr (2019, 2023) demonstrate that multi-level clustering could make a big difference. Accordingly, we experiment with two alternative clusterings. Specifically, to obtain our main results, we follow most of the existing literature and cluster the standard errors by country pair (i.e. $\text{Cov}[\varepsilon_{ij,t}, \varepsilon_{ij,d}] \neq 0$ for all t, d , and it is zero otherwise). In addition, we also obtain estimates with three-way clustering (by exporter, importer and year, i.e. $\text{Cov}[\varepsilon_{ij,t}, \varepsilon_{il,d}] \neq 0$, $\text{Cov}[\varepsilon_{ij,t}, \varepsilon_{kj,d}] \neq 0$, $\text{Cov}[\varepsilon_{ij,t}, \varepsilon_{kl,t}] \neq 0$, and these terms are zero otherwise). The implication for our command is that it should include an option for alternative clusterings.

Before we continue, we note that our econometric model (1) can be estimated at any desired level of aggregation, that is, product versus industry versus sector versus aggregate. For expositional simplicity, we do not include any notation to denote products, sectors, and so on. However, since the disaggregated gravity equation has solid theoretical foundations (e.g. Anderson and van Wincoop (2004) on the demand side, and Costinot *et al.* (2012) on the supply side), we can rely on clear guidance for some aspect of the disaggregated gravity estimations with pooled data; for example, the multilateral resistances should be of dimensions exporter-product-time and importer-product-time.

As a result, the corresponding three sets of fixed effects in our model would become exporter-product-time ($\phi_{i,t}^k$), importer-product-time ($\psi_{j,t}^k$) and country-pair-product ($\vec{\gamma}_{ij}$), where the superscript k is used to denote the product level (or any other level of aggregation). In addition, some bilateral policies (e.g. sanctions) may be implemented at the product level too. Moreover, regardless of whether a given policy, including EU membership, is implemented at the disaggregated or aggregate level, it is very likely that the corresponding effects on trade would be heterogeneous across products, sectors, and so on. Finally, the disaggregated dimension should also affect the clustering in our estimations. The main implication for our command is that it should be able to handle even more fixed effects and a larger number of gravity covariates.

3 | DID WITH ETWFE AND HIGH-DIMENSIONAL FIXED EFFECTS

This section describes a new (Stata) command that can be used to estimate staggered DiD models using an ETWFE approach while allowing for high-dimensional fixed effects (`jwddid`).¹⁴ To demonstrate the usefulness of the new command, we rely on it in subsequent sections to obtain all estimation and post-estimation results and figures in the paper. To build the command, we capitalize on three influential strands of the literature. First, we take into account many recent contributions from the vibrant DiD econometrics literature (e.g. Hull 2018; de Chaisemartin and D’Haultfœuille 2020, 2023; Callaway and Sant’Anna 2021; Goodman-Bacon 2021; Sun and Abraham 2021; Wooldridge 2021, 2023; Borusyak *et al.* 2024; Goldsmith-Pinkham *et al.* 2024).

Second, we take advantage of the recent developments for fast computation and convergence of linear and non-linear models with high-dimensional fixed effects (e.g. Correia 2016b; Correia *et al.* 2020). Third, we rely on the empirical and theoretical trade gravity literature (e.g. Eaton and Kortum 2002; Anderson and van Wincoop 2003; Arkolakis *et al.* 2012; Yotov *et al.* 2016; Nagen-gast and Yotov 2025). The result from synthesizing the developments from these three strands of the literature is a fast, flexible and general estimation command—`jwddid`. As we demonstrate below, our command can easily accommodate all established methods for estimating trade gravity models. In addition, the command is quite general as it can be applied directly to gravity models for any bilateral flows, and even more broadly to any two-dimensional DiD model.

3.1 | The baseline model and estimation command

The analysis in this paper is based on the bilateral gravity model of trade. However, in order to highlight the generality of the command and link it more directly to the recent DiD literature, we introduce it and describe its features based on the following ETWFE estimator proposed by Wooldridge (2021):

$$Y_{i,t} = \alpha + \sum_{g \in G} \sum_{s=g}^T \theta_{g,s} D_{g,s} + \xi_i + \xi_t + \varepsilon_{i,t}, \quad (2)$$

where $Y_{i,t}$ is the dependent variable, $D_{g,t}$ is a dummy that takes value 1 if the observation is in the treatment group g on period t , and 0 otherwise, G is a set that indicates at what time treatment started for all units, T is the last period of the analysis, and ξ_i and ξ_t are sets of fixed effects for the individual and time dimensions, respectively.¹⁵ In this setup, the $\theta_{g,t}$ coefficients represent the average treatment effect that the treatment group g experiences at time t (i.e. $ATT(g, t)$, where ATT stands for average treatment effect on the treated). As described in Wooldridge (2021), allowing for a flexible specification of $\theta_{g,t}$ avoids the problems of bad controls and negative weights that have been identified in the literature as potential problems in the estimation of DiD models using traditional TWFE estimators. In principle, equation (2) can be viewed as an event-study specification that allows distinct event-time coefficients for each cohort. For example, for the 2007 cohort (the entry of Bulgaria and Romania into the EU), identification comes from the evolution of trade between Bulgaria and Romania and existing EU partners (including the bilateral trade between Bulgaria and Romania) relative to that of other country pairs, and relative to the baseline period before accession, while conditioning on exporter-year and importer-year fixed effects (i.e. multilateral resistances).

Based on equation (2), and accommodating a series of recommendations and developments from the related literature, the benchmark syntax for the proposed `jwddid` command, including its main options, is as follows:

```
jwddid y, ivar(i) tvar(t) gvar(g) method() never exovar(x_ex)
xtvar(x_t) /// xgvar(x_g) fevar() hettype(option) cluster(cvar)
```

Here, y is the dependent variable (e.g. bilateral trade in our case), `ivar(i)` is used to identify the individual panel data dimension or unit (e.g. a country-pair identifier in the gravity settings).¹⁶ Also, `tvar(t)` identifies the time dimension (e.g. the year in our case), and `gvar(g)` identifies the treatment cohorts (e.g. country pairs between EU members). Specifically, for observation it , g would take value zero if the unit is never treated (within the window of the analysis), and would take a value different from zero to indicate the year that treatment started for unit i . Following standard assumptions, this specification assumes that the treatment is an absorbing state; that is,

once a unit is treated, it remains treated for the rest of the analysis. If the treatment can switch off—thereby violating the absorbing-state assumption—then extensions that redefine cohorts to accommodate entry and exit, and allow for heterogeneous effects, are available (Wooldridge 2021, 2023). For example, this approach has been applied in the gravity setting to the case of sanctions with entry and exit (Kwon *et al.* 2024).

As described in Wooldridge (2023), the standard ETWFE model from equation (2) identifies the average treatment effect by imposing a linear parallel trends assumption. However, such an assumption may not be valid in some cases, such as when the dependent variable follows some limited distribution. Roth and Sant'Anna (2023) discuss a similar problem, stating that the choice of transformation of the dependent variable is crucial for the identification of the average treatment effect, and only under certain conditions would the ATT be identified for any transformation. This is particularly important in gravity settings, where, as discussed earlier, the non-linear PPML estimator is the established estimator.

Wooldridge (2023) addresses this issue by demonstrating that the linear ETWFE models can be adapted to allow for non-linear models, by simply imposing the linear parallel trends assumption only on the latent variable of the model, but not on the outcome itself. Capitalizing on that, the `jwddid` command allows the user to specify an estimator of choice using the `method()` option. By default, `jwddid` will use the `reghdfe` command of Correia (2016a; however, there are no restrictions on the type of `method`. Thus, for example, to estimate gravity models with PPML and high-dimensional fixed effects, as we do here, one can use the `ppmlhdfe` command of Correia *et al.* (2020), that is, `method(ppmlhdfe)`.¹⁷

Specification (2) makes the implicit assumption that parallel trends are satisfied, using all never treated and not-yet treated observations as controls (not included category) for the identification of treatment effects. If one instead wants to relax this assumption, then one can specify the option `never`. In this case, the only observations that are used as controls are the ones that were never treated. In principle, this is the same as the strategy proposed by Sun and Abraham (2021), allowing for full heterogeneity across all groups and all event time periods. Such a specification is also numerically identical to the one proposed by Callaway and Sant'Anna (2021) for the case where there are no covariates.

As described in Wooldridge (2021), it is possible to include covariates in the model, by simply adding corrections that enable us to easily identify the average treatment effect. However, following the literature on DiD models, the implicit assumption is that covariates are time-invariant. Command `jwddid` allows for adding covariates directly to the benchmark syntax, immediately after the dependent variable, and it does not impose any assumption on the covariates, but the user should be aware of the implications. If one uses the option `xasis`, then the command will use the covariates without demeaning them, which may save some computation time.

The default option when introducing additional covariates is to interact all of them (or the demeaned transformations) with the same level of covariate heterogeneity. Sometimes, however, one may not be interested in estimating the same level of heterogeneity for all covariates. It may be possible, for example, to consider separate sets of covariates that could be interacted only with the time or group dimensions. Specifically, assume that there are no variables that we wish to consider for the treatment heterogeneity, but instead consider three sets of covariates: variables without further interactions (x^{EX}), variables that would be interacted with the time variables only (x^T), and variables that would be interacted with group indicators only (x^G). Command `jwddid` can accommodate such scenarios with the options `exovar(x_ex)`, `xtvar(x_t)` and `xgvar(x_g)`, respectively.

Sometimes, especially in gravity settings such as ours, it may be beneficial to include high-dimensional fixed effects (and interactions with fixed effects) that are different from the individual and time fixed effects. It is possible to request the inclusion of those types of fixed effects using the option `fevar()`, which is valid only if one is using the default estimator `method(reghdfe)` or `method(ppmlhdfe)`. In both cases, the additional fixed effects (or interactions) are included

in the estimation of the model without further interactions. This is particularly convenient for gravity-type estimations, which often require the use of source-time and destination-time fixed effects.

The number of estimated parameters in equation (2) can grow quickly with the number of groups/cohorts, time periods and covariates. This could lead to increasing computational burden of the estimation. An alternative, which is already implemented via `xthdidregress` and `hdidregress` in Stata 18, is to estimate a model that reduces the heterogeneity of the treatment effects. Specifically, it allows treatment effects to vary across cohorts, across absolute time, or across relative time. The `jwddid` command allows the user to specify/impose restrictions along each of these dimensions with `hettype(option)`, where `option` can be `time`, `cohort` or `event`. If no option is selected, then the command will estimate the model by allowing for full cohort-time heterogeneity. For convenience, `twfe` can also be specified, which imposes treatment effect homogeneity across time and cohorts, that is, the TWFE specification.

Finally, by default, the `jwddid` command assumes clustered standard errors at the `i` level (i.e. at the country-pair dimension in our case), which is consistent with the standard approach in the gravity literature. However, if a different level of clustering is desired, for example, at the level of a variable '`cvar`', then the user can specify the `cluster(cvar)` option. The `cluster(cvar)` option is not required, but can be used to request the standard errors to be clustered at the level `cvar` when using `reghdfe` or `ppmlhdfe`. This is useful for gravity estimations where sometimes three-way clustering (e.g. by source, destination and time) may be recommended (Egger and Tarlea 2015).

3.2 | Post-estimation options and analysis

After the estimation of the model, under the default options, one can recover directly the estimates of the group- and time-specific ATTs. However, one may also be interested in obtaining aggregated ATTs for the overall data, across groups, or periods, or dynamic effects. Furthermore, when the underlying method is a non-linear model, the estimated coefficients cannot be directly interpreted as the average treatment effect on the outcome, but only on the latent variable. To accommodate such needs, the `jwddid` command comes along with the post-estimation command `jwddid_estat/estat` or just `estat`:

```
estat [aggregation] [pw = weight],
      [vce(unconditional)] [margins_options]
```

Command `estat` can implement four types of 'aggregations', including (i) an ATT for all observations that were treated at some point in time (`estat simple`), (ii) an average treatment effect for observations that were treated at time `g` (`estat group`), (iii) an average treatment effect at time `t` for all observations that were effectively treated at that point (`estat calendar`), and (iv) dynamic treatment effects, also known as event studies (`estat event`).

The default option for the estimation of the aggregated ATTs is to use the weights that were used in the estimation of the model. However, if the user wants to use different weights, it is possible to do so using the standard `[pw = weight]` option.

Wooldridge (2021) suggests that when one estimates standard errors for the aggregated ATTs, one should use the `vce(unconditional)` option in Stata, to allow for uncertainty in the explanatory variables. But `jwddid_estat/estat` does not use this approach by default, because it requires that the underlying command is able to produce scores for the estimated model. For example, if the model is estimated using `method(regress)`, then the scores will be available, and unconditional standard errors for the aggregated ATTs can be estimated. However, this is not possible if one uses `reghdfe` or `ppmlhdfe`.

In addition, standard margins options can be used after the post-estimation command `estat`. In particular, given our focus on the gravity setting and the `ppmlhdfc` method, we use the option `predict(xb)`. This yields (approximate) proportional treatment effects (Wooldridge 2023; Nagengast and Yotov 2025). Because of this, standard errors are typically estimated using the delta method. If using methods such as `regress` or `logit`, one can also request `vce(unconditional)` so standard errors also account for sampling variation of covariates.

As described earlier, the default aggregation considers all treated observations, imposing restrictions only in terms of time, group or event dimensions. However, one may be interested in imposing further restrictions that could leverage on the use of covariates, such as a dummy variable `dx`, in the model specification. As usual, one can request the estimation of the aggregated ATTs for the whole sample. Alternatively, one can also impose the added restriction that the covariate `dx` is 0 or 1, using the option `orestriction()`, that is, `orestriction(dx==1)`. The expression inside the parentheses should be a valid Stata expression that is used when calculating the aggregated ATTs.

After aggregate effects have been estimated, the user may want to store the results for further analysis or reporting. Because `estat` uses margins in the background, the default option is to store the results in memory as `r()` elements. Alternatively, `jwddid_estat/estat` allows the user to store the outputs of the command using three different options: (i) store them as the current estimations in memory `e()` (using the option `post`); (ii) store them in memory as `name` (using the option `estore(name)`); (iii) store them in a file `filename`, as a `ster` file (using the option `esave(filename)`).

After time, group or event aggregations are estimated (or simple with the `over()` option), it is possible to request plotting those results using `estat plot`. The basic syntax is to type it after the aggregation command:

```
estat plot, style(style) twoway_options
```

where the option `style` allows the user to select the style of the plot, and standard `twoway` graph options can be used after the `estat plot` command. The `plot` command allows for a lot of further flexibility with only minimal command-specific options, which we describe in detail in Subsection A.1 of the Online Appendix, where we also offer further details on the `jwddid` command and its options. We also provide specific examples for the implementation of the command in our gravity setting in Section 5.

4 | DATA

We rely on standard/established data sources to compile several estimating samples for our empirical analysis. Our largest estimating sample is an unbalanced panel dataset of aggregate bilateral trade flows at the country-pair level, which covers 260 countries over the period 1950–2019. In addition to all goods trade flows, in Subsection 6.1, we also obtain results with data for manufacturing trade, which covers 229 countries over the period 1980–2016. We also use data on several gravity variables. Finally, we note that to limit the influence of other events on the estimates, and to ensure the same number of pre-treatment observations for all cohorts, we restrict the estimating samples to six years before EU accession for treated country pairs.

4.1 | Data sources

4.1.1 | Aggregate trade flows

The data on nominal aggregate bilateral trade flows were constructed by Felbermayr *et al.* (2020), who combined the data on aggregate bilateral trade flows at the country-pair level from the two

most comprehensive and most widely used trade datasets: (i) the DoTS of the IMF, and (ii) the United Nations COMTRADE database.¹⁸ To obtain the maximum number of trade flow observations, Felbermayr *et al.* (2020) use a mirroring procedure. Specifically, consistent with trade theory, where trade volumes are measured at delivered prices (e.g. Eaton and Kortum 2002; Anderson and van Wincoop 2003, 2004), cost, insurance and freight (CIF) imports are used as the baseline data. Then missing values are replaced with data on free on board (FOB) exports.

The trade data of Felbermayr *et al.* (2020) cover the period 1950–2019, and their main advantage for our purposes is that they cover a very long period of time. To obtain our main estimates, we rely on a subsample of this sample, which covers 98% of bilateral trade flows in the world; however, we also demonstrate that our results are robust to using a medium-size sample, which covers 99% of trade flows, and also the full sample. We also experiment with several variants of the dataset to test the robustness of our results to the treatment of missing values in the IMF DoTS dataset.

4.1.2 | Manufacturing trade flows

The manufacturing trade flows data come from the SGD of the WTO (Larch *et al.* 2025a).¹⁹ The SGD includes aggregate manufacturing trade data for 229 countries over the period 1980–2016, and it has three advantages for our purposes. First, as noted earlier, according to Eurostat (2023), four-fifths of all intra-EU trade is in manufacturing goods. Thus this is by far the most important sector for intra-EU trade. Second, on a related note, the sectoral estimates for manufacturing will complement our aggregate results. Finally, the SGD includes consistently constructed domestic trade flows, which will enable us to obtain and compare estimates of the EU effects with and without domestic trade flows.

4.1.3 | Other data

We also coded the data on EU membership according to Figure 2, and we use data on several standard gravity variables in our analysis, which come from the Dynamic Gravity Dataset (DGD) of the US International Trade Commission (Gurevich and Herman 2018).²⁰ There are various reasons to expect that the onset of the EU effects on trade would precede the date of the official EU accession. For example, accession to the EU is a multi-stage process with a gradually increasing degree of commitment. Specifically, the process involves six steps, including (i) application for accession, (ii) opinion from the Commission, (iii) granting of candidacy, (iv) accession negotiations, (v) signing of an accession treaty, and (vi) entering into force of the accession treaty (Leppert 2022). It is quite possible to observe certain adjustments in trade costs as well as in the behaviour of economic agents at each stage of the accession process. However, we believe that one is most likely to observe such changes during the period between the signing of the accession treaty and its official entry into force. This period varies by cohort, and it is usually between about one year (e.g. for the 2004 cohort) and about two years (e.g. for the 2007 cohort).

Against this backdrop, for our main analysis, we will assume an ‘onset’ of the EU effects two years before the official entry comes into force. In addition, in the robustness analysis, we consider two alternative strategies: (i) we omit periods with anticipation effects; and (ii) we use the year of EU accession as the treatment onset.

4.2 | Descriptive statistics

Figure 2 shows all countries that were, at some point, part of the EU, by accession year.²¹ It shows that EU enlargement occurred in several waves, usually with several countries entering

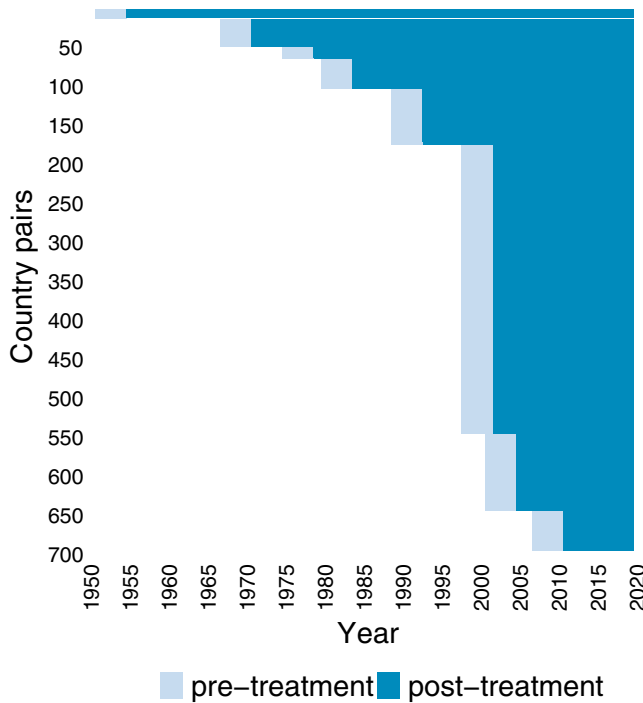


FIGURE 3 Cohorts and treatment status over time. *Notes:* This figure shows the treatment status of treated country pairs over time. Pre-treatment years are coloured in light blue. Post-treatment years are coloured in dark blue. Note that, as explained in the text, the treatment onset was shifted forward by two years to capture anticipation facts. As a result, for example, the post-treatment years for the 1957 cohort already begin in 1955. The figure was generated using the Stata module `panelview` (Mou and Xu 2022).

the EU at the same time. Note that separate foreign trade data for Belgium and Luxembourg is available only from 1999 onwards. To avoid breaks, we consider Belgium–Luxembourg as one exporter/importer throughout the analysis. Note also that data for Belgium–Luxembourg are available only in our dataset from 1960 onwards, such that Belgium–Luxembourg is not included in the 1957 cohort.

Figure 3 shows the observations of country pairs in the treatment group by pre- and post-treatment status over time. The treatment cohorts are of different sizes—in accordance with the different waves of EU enlargement from Figure 2—and also of different lengths, depending on the year of treatment onset. Note that country pairs are combinations of exporters and importers. Therefore cohorts are composed of not just new members, but also combinations of new and old members, an aspect that we analyse in detail in Subsection 5.3.

For completeness, Table 1 reports the number of observations, country pairs, exporters, importers and years for different groups in the dataset. Note that the disparity in the numbers of not-yet-treated and treated (i.e. post-treatment) observations reflects our sample construction: the former is limited to a fixed four-year pre-treatment window per treated country pair (we include six years before accession, and shift the onset forward two years to capture anticipation), whereas the latter is uncapped, so early-treated pairs contribute many post-treatment observations. Table 2 reports the average of the variables Distance (in kilometres), Contiguity, Language and Colony for different groups in the dataset. Note that, not surprisingly, treated country pairs differ quite substantially with regard to standard gravity variables. Country pairs in the EU are closer together, and more likely to share a common border, but less likely to share a common language or a past colonial relationship. The most salient difference between treated

TABLE 1 Descriptive statistics: observations along different dimensions.

Group	Observations	Pairs	Exporters	Importers	Years
1957 cohort	780	12	4	4	65
1973 cohort	1764	36	8	8	49
1981 cohort	656	16	9	9	41
1986 cohort	1368	38	11	11	36
1995 cohort	1944	72	14	14	27
2004 cohort	6660	370	24	24	18
2007 cohort	1470	98	26	26	15
2013 cohort	468	52	27	27	9
Treated	15,110	694	27	27	65
Not-yet treated	2776	694	27	27	31
Never treated	399,245	7232	90	90	70

Notes: The table reports the number of observations, country pairs, exporters, importers and years for different groups in the baseline estimation sample from the ETWFE estimate in column (2) of Table 3. ‘Cohort’ refers to all post-treatment observations of new EU country pairs in a particular year. ‘Treated’ refers to all post-treatment observations of all cohorts. ‘Not-yet treated’ refers to all pre-treatment years of all cohorts. ‘Never treated’ refers to all observations of country pairs that did not enter the EU during the sample period.

TABLE 2 Descriptive statistics: summary statistics of covariates for different groups.

Group	Distance	Contiguity	Language	Colony
1957 cohort	926	0.50	0.33	0.00
1973 cohort	991	0.11	0.33	0.06
1981 cohort	2075	0.00	0.75	0.00
1986 cohort	1702	0.11	0.00	0.00
1995 cohort	1617	0.08	0.14	0.00
2004 cohort	1503	0.08	0.14	0.01
2007 cohort	1473	0.06	0.16	0.00
2013 cohort	1124	0.08	0.38	0.00
Treated	1458	0.10	0.19	0.01
Never treated	7945	0.02	0.34	0.03

Notes: The table reports the average of the variables Distance (in kilometres), Contiguity, Language and Colony for different groups in the baseline estimation sample from the ETWFE estimate in column (2) of Table 3. ‘Cohort’ refers to all new EU country pairs in a particular year. ‘Treated’ refers to all cohorts. ‘Never treated’ refers to all country pairs that did not enter the EU during the sample period.

and never-treated country pairs is the substantially smaller distance between EU member states. In Section 6, for robustness, we consider an extension of the ETWFE model relaxing the parallel trends assumption, which now only needs to hold conditional on distance.

5 | ESTIMATES OF THE EU EFFECTS ON TRADE

This section presents our main results. In Subsection 5.1, we test the identifying assumptions of the ETWFE model. In Subsection 5.2, we obtain and discuss our main results. Finally, in Subsection 5.3, we explore the heterogeneity of the EU effects across several dimensions. Importantly, all results and all figures that we present in this section are obtained with the new

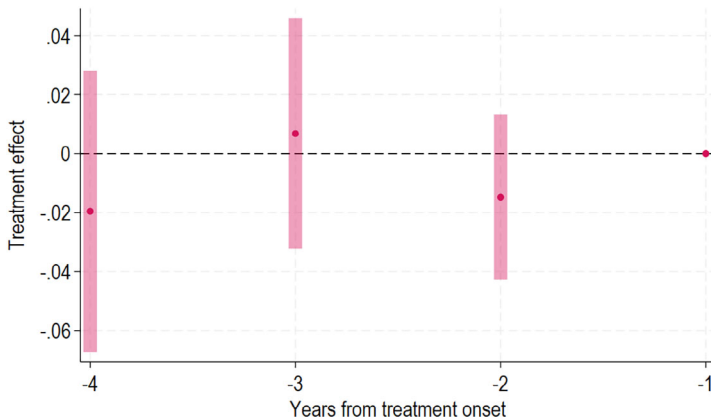


FIGURE 4 Pre-treatment effects. *Notes:* The figure reports pre-trend estimates from a PPML estimation of equation (1) augmented with cohort-year-specific placebo effects in the four years before treatment onset. To avoid misleading visual inferences, we follow the recommendation by Roth (2024) and put our pre-treatment estimates in a different plot from the post-treatment estimates. The regression is estimated using the full sample following Wooldridge (2021, 2023). The cohort-year-specific treatment effects were aggregated across cohorts to obtain event-time-specific treatment effect estimates. The 95% confidence intervals are shown using standard errors clustered by country pair. *Stata code:* `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci idt_cj) never estat event, predict(xb) pretrend window(-4 -1) estat plot, ytitle("Treatment effect") xtitle("Years from treatment onset") pstyle1(p2)`

estimation command. To highlight this, we have included the command syntax used to obtain each table and figure in this section in the notes for the corresponding exhibit. In this regard, we define `trade` to be exports, which vary over the exporter–importer–year dimension, `id_ci_cj` is a country-pair ID, `year` indicates the year in the dataset, `FT_EU` indicates the first year in which both the exporter and importer of a country pair were part of the EU, and `idt_ci` and `idt_cj` are exporter-time and importer-time identifiers.

5.1 | Test of identifying assumptions

In this subsection, we test the no anticipation assumption and the parallel trends assumption of the ETWFE model (Wooldridge 2021, 2023; Nagengast and Yotov 2025). We do so by following Wooldridge (2021, 2023) by augmenting the estimating equation (1) with cohort-year-specific placebo effects in the years before treatment onset.²² Figure 4 shows the resulting placebo treatment effects aggregated across cohorts by event time. The results provide evidence against the violation of the identifying assumptions before EU accession. The pre-treatment effects are statistically insignificant in the years before treatment onset.²³ We therefore proceed with our main analysis.

5.2 | Main estimation results

Our main aggregate EU estimate appears in column (2) of Table 3, obtained from an estimating sample that covers 98% of world trade. The main implication of these results is that on average, membership in the EU has led to a significant increase in bilateral trade flows among its member states. Specifically, the EU estimate in column (2) implies that the EU has led to an increase of more than 52% in the bilateral trade flows between its member states.²⁴

TABLE 3 Average EU effects on trade: TWFE versus ETWFE for different samples.

	TWFE (1)	ETWFE (2)	TWFE (3)	ETWFE (4)	TWFE (5)	ETWFE (6)
EU_{ijt}	0.332*** (0.035)	0.422*** (0.037)	0.337*** (0.035)	0.435*** (0.037)	0.338*** (0.035)	0.431*** (0.036)
Sample	Baseline	Baseline	Medium	Medium	Large	Large
Observations	417,131	417,131	642,570	642,570	2,303,571	2,303,571
Exporters	90	90	112	112	260	260
Importers	90	90	112	112	260	260
Years	70	70	70	70	70	70
Coefficients	1	260	1	260	1	260
p -value H_0 : TWFE = ETWFE		0.0014		0.0004		0.0005
Exporter $\times$ importer FE	Yes	Yes	Yes	Yes	Yes	Yes
Exporter $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes
Importer $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table presents PPML regression results using a TWFE model, in which the cohort-year-specific effects in equation (1) were substituted with a homogeneous EU effect, and the ETWFE model (equation (1)), for which the cohort-year-specific treatment effects were aggregated to obtain an aggregate treatment effect estimate. The dependent variable is exports, which vary over the exporter–importer–year dimension. The ‘Small’ sample, which is used to obtain the ‘Baseline’ results in columns (1) and (2), contains 90 countries, accounting for 98% of world exports. The ‘Medium’ sample, which is used to obtain the results in columns (3) and (4), contains 112 countries, accounting for 99% of world exports. The ‘Large’ sample, which is used to obtain the results in columns (5) and (6), contains the full set of countries from our dataset. ‘Coefficients’ reports the number of estimated coefficients apart from the fixed effects. Standard errors in parentheses are clustered by country pair. The row labelled ‘ p -value H_0 : TWFE = ETWFE’ reports the results of a standard F -test of the equality of the corresponding TWFE and ETWFE coefficients, where the covariance between the two coefficients is accounted for using a seemingly unrelated regression specification.

***, **, * indicate significance at the 1%, 5%, 10% level, respectively.

Stata code: `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdf) fevar(idt_ci idt_cj) /// hettype(twfe) // Columns (1), (3) and (5) jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdf) fevar(idt_ci idt_cj) // Columns (2), (4) and (6) estat simple, predict(xb) // All columns`

While large and statistically significant, our estimate of the impact of the EU is smaller than the corresponding estimates from Mayer *et al.* (2019), who find that the single market version of the EU has tripled trade among member countries. We see several possible explanations for the differences. First, the main results in Mayer *et al.* (2019) seem to have been obtained with the OLS estimator, while we rely on PPML.²⁵ Second, the lengths of the estimating samples are different; for example, the long sample of Mayer *et al.* (2019) ends in 2012, while ours ends in 2019. Thus we are probably capturing the smaller effects on the new members. Third, we do not distinguish between the effects of the European Community versus the single market, and from a methodological perspective, our non-linear methods do not allow for multiple treatments. Finally, we note that our estimate is significantly larger than recent estimates of the effects of trade agreements (e.g. Baier *et al.* 2019; Larch and Yotov 2024; Nagengast and Yotov 2025). Moreover, it is also very likely that the average estimates in Table 3 may be masking significant heterogeneity across various dimensions—for example, over time, across cohorts, and across EU member states. In subsequent analyses, we demonstrate that this is indeed the case.

To highlight the policy and methodological implications of our ETWFE estimates, we compare them to those from a standard TWFE gravity model, which are reported in column (1) of Table 3. While the difference between the estimates does not seem to be very large, they are statistically significantly different from each other. Moreover, the difference between the implied trade volume effects between the two specifications is 13 percentage points (39.4% from the TWFE model versus 52.5% from the TWFE specification)—that is, not negligible. To learn more about

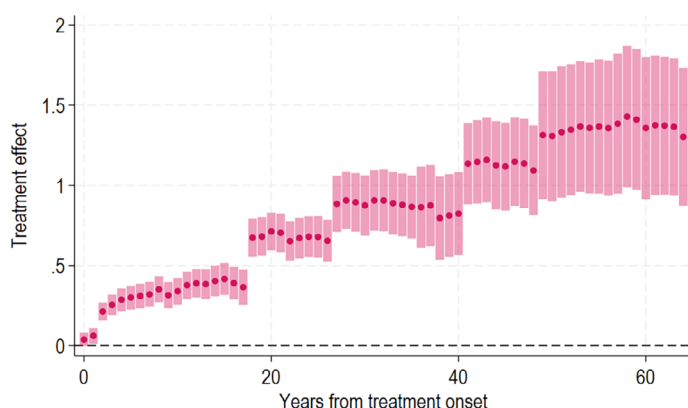


FIGURE 5 Event-time-specific EU effects. *Notes:* The figure reports event-time-specific treatment effects, for which the cohort-year effects (corresponding to the aggregate EU effect in column (2) of Table 3) were aggregated across cohorts. Note that the treatment onset was shifted forward by two years (see Subsection 4.1) such that the year of the EU accession corresponds to time period 2. The 95% confidence intervals are shown using standard errors clustered by country pair. *Stata code:* `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci idt_cj) estat event, predict(xb) estat plot, ytitle("Treatment effect" xtitle("Years from treatment onset"))`

the reasons behind the differences between the TWFE and ETWFE results, a decomposition analysis of the *OLS* TWFE estimate in the spirit of de Chaisemartin and D'Haultfœuille (2020) can be performed. The underlying assumption is that the *OLS* decomposition results are likely informative about the PPML estimates as well. Given that the pattern of differences between the TWFE and ETWFE results is relatively similar for *OLS* and PPML (see Table 5 below), we hypothesize that this is indeed the case. The *OLS* TWFE decomposition results suggest that the TWFE are likely downward biased since they place larger weights on short-term effects and on the effect of late cohorts, which are both associated with smaller treatment effects. We confirm the robustness of our main result in columns (3) and (4) of Table 3, where we use a medium sample that covers 99% of trade flows in the world, and in columns (5) and (6), where we use the full sample. The EU estimates are very similar across all ETWFE specifications, and in each case, they are also larger than the corresponding TWFE results.

We complement the aggregated EU estimates from Table 3 with a set of event-time-specific ETWFE EU effects, which are visualized in Figure 5. Several patterns stand out from the evolution of the EU effects over time. First, in line with the literature and the discussion in Subsection 4.1, we find positive and significant anticipation effects in the first two years before EU accession. While the anticipation effects are statistically significant, they are at the same time substantially smaller than the effects after the actual EU accession (i.e. from period 2 onwards in Figure 5). Second, consistent with our aggregate estimates, all estimates of the phasing-in effects of the EU on trade are positive and statistically significant. Third, the EU effects increase over time throughout the period of investigation. A possible interpretation of these results is that the EU effects are long-lasting. However, it is also possible that the average event-time-specific effects hide certain cohort-specific patterns in the evolution of the EU effects. The few ‘jumps’ that we see in Figure 5 are consistent with this hypothesis. As we will see in the next subsection, those ‘jumps’ are due to composition effects, because the underlying event-study plots by cohort are smooth and have no ‘jumps’. In particular, coefficients for early post-accession periods are estimated using both early and late cohorts, whereas later event times reflect only the early cohorts, since late cohorts by construction lack sufficiently long post-treatment histories. This changing composition of contributing cohorts over time generates the apparent shifts in the aggregated event-time estimates.

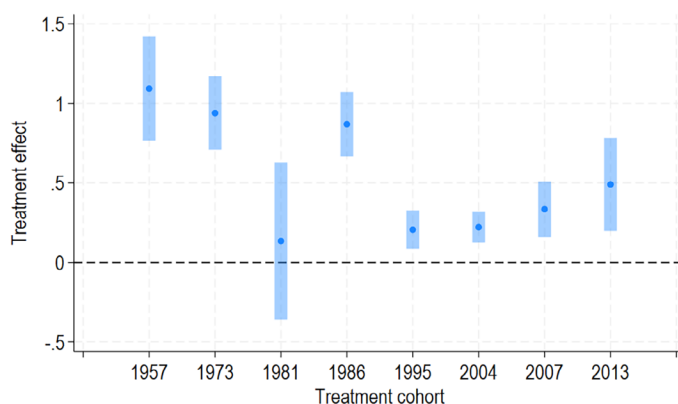


FIGURE 6 Cohort-specific treatment effects. *Notes:* The figure reports cohort-specific treatment effects, for which the cohort-year effects (corresponding to the aggregate EU effect in column (2) of Table 3) were aggregated across event time. The 95% confidence intervals are shown using standard errors clustered by country pair. *Stata code:* `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdf) fevar(idt_ci idt_cj) estat group, predict(xb) estat plot, ytitle("Treatment effect") xtitle("Treatment cohort")`

5.3 | Disaggregated EU effects on trade

Motivated by the results in the previous subsection, and in order to demonstrate the importance of the heterogeneity-robust DiD approach for estimating the EU effects, in this subsection, we explore several dimensions of the underlying heterogeneity in the EU effects. In particular, we consider heterogeneity across cohorts, cohort-specific event-study results, and we also obtain directional estimates on trade between ‘old’ and ‘new’ EU members depending on the recency of their EU membership.

5.3.1 | EU effects by cohort

We start with an analysis by cohort. In this regard, the heterogeneous waves of EU accession (see Figure 2) offer a very interesting setting. Our estimates are visualized in Figure 6. Two main results stand out.

First, all cohort estimates are positive, and all but one of them are statistically significant. Thus, consistent with our main findings, the implication is that the EU has benefited its members’ exports. The single exception is the 1981 cohort, which only consists of all country pairs of previous EU members with Greece.²⁶

The second conclusion that we draw based on the results in Figure 6 is that the effects of the EU have been quite heterogeneous across cohorts. The results in Figure 6 suggest that apart from the 1981 cohort, the earlier cohorts—that is, the 1957 cohort (country pairs consisting of the founders), the 1973 cohort (country pairs consisting of previous EU members with Denmark, Ireland and the UK) and the 1983 cohort (country pairs consisting of previous EU members with Portugal and Spain)—have benefited more from the corresponding EU enlargements.

A possible explanation for this is that the benefits from the most recent rounds of enlargements have not materialized yet; that is, the differences that we capture can be due to composition effects because later cohorts have fewer long-term effects than early cohorts. However, the overall pattern of results remains qualitatively similar—apart from a very strong effect for the 2013 cohort comparable to effects for the 1973 and 1986 cohorts—when conditioning on the same number of years for every cohort (see Figure A2 in the Online Appendix).

A potential caveat is that our cohort estimates rely on trade data that do not extend far enough back to cleanly isolate the impacts of earlier trade agreements and integration frameworks (such as the European Free Trade Association (EFTA), the Central European Free Trade Agreement (CEFTA), and bilateral Association Agreements). For cohorts that participated in these precursor agreements, the discrete jump that we attribute to formal EU accession may be mechanically attenuated, whereas cohorts without such precursors could exhibit a larger increase. Consequently, we cannot exclude that part of the heterogeneity that we document across cohorts may have arisen from this mechanical artefact rather than from genuine differences in accession benefits. Disentangling these layers will require harmonized, longer-span trade series extending further back, which is an interesting avenue for future research.

Subsequently, we explore whether differences in phasing-in effects might leave room to expect additional trade effects for late cohorts in the years after 2019.

5.3.2 | Event-study results by cohort

In our next experiment, we obtain event-study results by cohort—that is, we trace the evolution of the EU effects over time for each of the cohorts in our sample. Our findings are reported in Figure 7, and overall, they demonstrate that the more aggregated estimates that we have discussed thus far indeed conceal significant heterogeneity. Several results are noteworthy.

First, we observe that all cohort-specific event-study plots are smooth over time, and that there are no ‘jumps’. However, note that estimates of event-study effects far away from treatment onset are available only for earlier cohorts. In combination with the heterogeneous EU effects that we observe across cohorts (see Figure 6), this results in the discontinuities that were apparent in the aggregate event-study plot (see Figure 5)—that is, the ‘jumps’ are due to composition effects.

Second, we see that the positive effects for the 1973 and 1986 cohorts accrue relatively quickly. We should also note that even if these effects (which last for 10–15 years) seem to be exhausted relatively quickly, they are still significantly more long-lasting than recent estimates of the duration of the phasing-in effects of free trade agreements (e.g. Egger *et al.* 2022). For the 1957 cohort, positive effects also appear quickly and then begin to rise again strongly around 5 years after the foundation of the European Community, perhaps reflecting that the benefits for the initial founders set in with a certain delay as the initial integration took more time than for later cohorts.

Third, we also see from Figure 7 that both the magnitude and the evolution of the EU effects vary significantly across cohorts. Thus, for example, the effects for the 1973 and 1986 cohorts are large and are exhausted relatively quickly, while the effects for the 1995 cohort are smaller but more long-lasting. There were hardly any anticipation effects for the 2004 cohort, while the remaining effects have been relatively immediate, small and stable over time. A possible caveat with the interpretation of the results for the 2004 cohort is that these estimates may be masking significant heterogeneity across the large number of countries that are in this cohort. The effects for the 2007 cohort (Bulgaria and Romania) are relatively large and gradual, and they seem to be exhausted already, about 10 years after accession. Finally, the impact of EU membership on Croatia’s trade (the single country in the 2013 cohort) has been strong, and it is ongoing 6 years after its accession. Therefore, from this perspective, the cohort heterogeneity observed in Figure 6 also cannot be explained by differences in phasing-in effects, while the final effect of the 2013 cohort might very well end up in the ballpark of the 1957, 1973 and 1986 cohorts.

5.3.3 | EU effects by recency of membership

We conclude with an exploration of possible directional asymmetries in the effects of the EU based on recency of membership. Specifically, we split the countries in each cohort into two

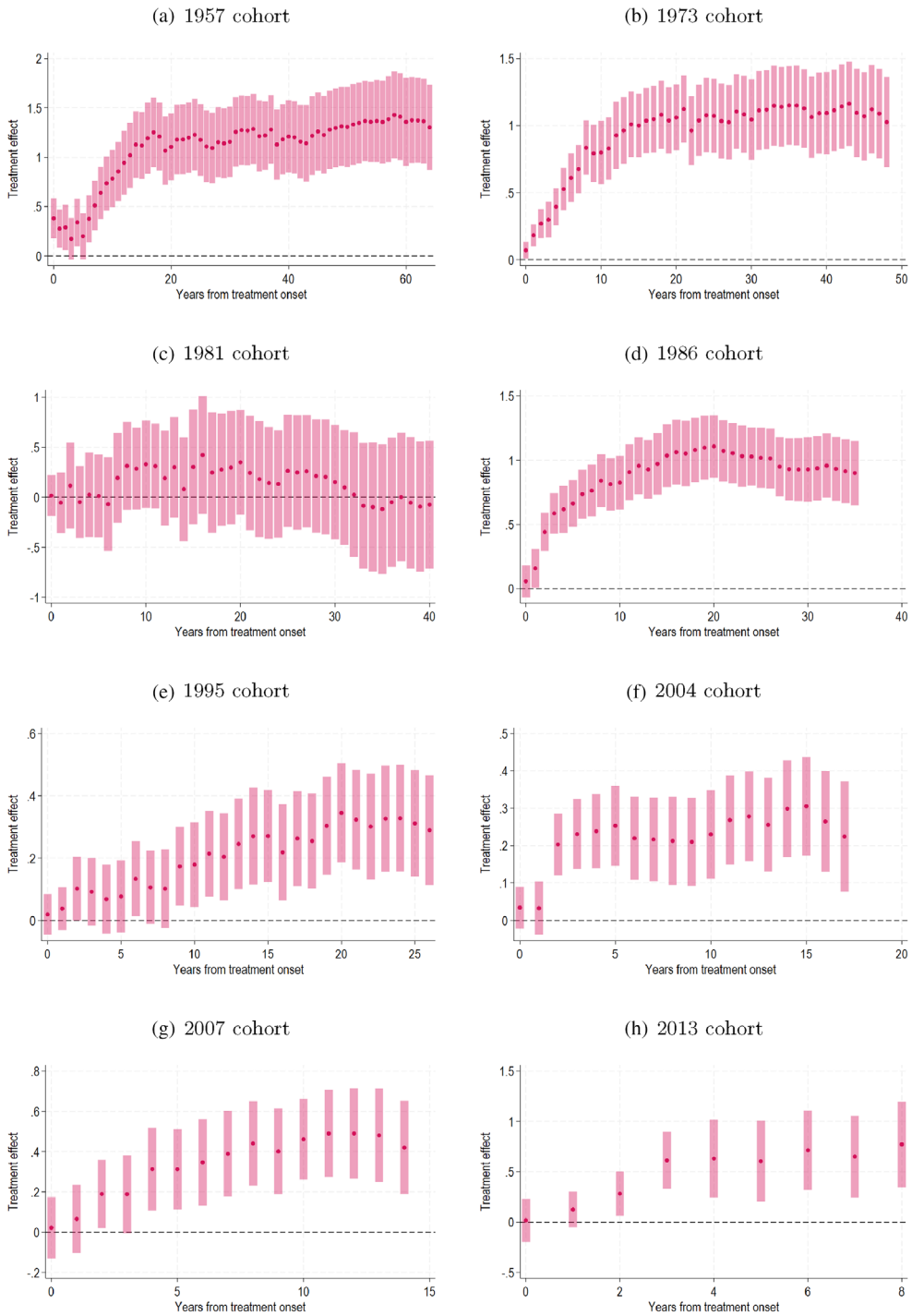


FIGURE 7 Event-time specific EU effects by cohort. *Notes:* The figure reports event-time-specific EU effects by cohort, i.e. all estimated cohort-year effects (corresponding to the aggregate EU effect in column (2) of Table 3). The 95% confidence intervals are shown using standard errors clustered by country pair. *Stata code:* `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdf) fevar(idt_ci idt_cj) estat event, predict(xb) ores(FT_EU==‘year’) // ‘year’ corresponds to the first post-treatment year of a cohort estat plot, ylabel(“Treatment effect”) xlabel(“Years from treatment onset”)`

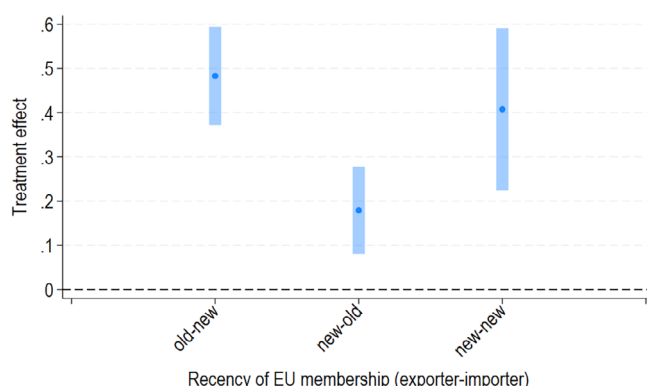


FIGURE 8 EU trade effects by recency of membership. *Notes:* The figure reports the average EU effect by recency of membership from a specification allowing for additional heterogeneity, for which the cohort-recency-year effects were aggregated across cohorts and event time. The 95% confidence intervals are shown using standard errors clustered by country pair. *Stata code:* `jwddid trade i.newold, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci idt_cj) estat simple, predict(xb) over(newold) // 'newold' indicates the direction of trade flows by recency estat plot, ytitle("Treatment effect", size(large)) xtitle("Recency of EU membership (exporter-importer)") /// xlabel(0 " " 1 "old-new" 2 "new-old" 3 "new-new" 4 " ", angle(45)) ylabel(0 (0.1) 0.6`

groups: 'old' members that were already part of the EU, and 'new' members that joined the EU in that particular year. Note that this definition is dynamic with regard to countries, and always relative to a particular cohort—that is, a country will initially be a 'new' member in the cohort of the year in which it entered the EU, but it will be an 'old' member in subsequent cohorts. Then we obtain estimates of the effects of the EU depending on the direction of trade flows, such as old-to-new, new-to-old and new-to-new. Note that by definition, we cannot estimate the direction old-to-old as these country pairs are already part of the other categories (in previous cohorts).

Our results appear in Figure 8, and we draw the following conclusions based on them. First, trade has increased for all groups and in each direction, and all estimates are statistically significant. Second, the effects have been asymmetric. Specifically, the estimate of the effects of the EU on the exports from old-to-new members (0.483, standard error 0.057) and from new-to-new members (0.408, standard error 0.094) is substantially larger (more than twice as large) than the corresponding estimate for the exports of new-to-old members (0.179, standard error 0.050). We note that this pattern also mostly holds across cohorts (see Figure A3 in the Online Appendix) and cannot be explained by differences in pre-trends (see Figure A4 in the Online Appendix). Furthermore, the differences between the new-to-old members and the other two categories seem to be particularly pronounced for the EU accession waves from 1995 onwards.

These asymmetries are consistent with the estimates from Anderson and Yotov (2022), who explore the EU impact on the extensive margin of trade, and also find significantly stronger impact on the number of products traded from 'old' to 'new' members than from 'new' to 'old' members (here defined at the country level). As noted in Anderson and Yotov (2022), a possible explanation for this result is that the new members have not been very successful in placing their products on the more competitive Western European market. An alternative explanation could be that the more recent accession waves have involved less developed (low-wage) countries that may have mostly exported cheaper and more labour-intensive intermediates.

Overall, the results in Figure 8 reinforce our previous conclusion that the EU has promoted trade among its members significantly. However, we now also see that the increase in exports may have been asymmetric and in favour of the 'old' member states. We do note that: (i) the new members have also enjoyed larger exports due to their EU membership, particularly with other

‘new’ member states; (ii) it is possible that their exports to old members will keep increasing over time, and that they will also further profit from future enlargements as by now seasoned (‘old’) EU members; and (iii) they have further benefited from their imports from the rest of the EU. Nevertheless, we believe that our results may have important policy implications.

6 | ROBUSTNESS ANALYSIS

This section describes a series of robustness checks, which are split into three groups: (i) different datasets (Subsection 6.1), (ii) degree of heterogeneity and alternative estimators (Subsection 6.2), and (iii) DiD- and ETWFE-specific assumptions (Subsection 6.3).

6.1 | Dataset

With regard to the dataset, first, we use the WTO’s SGD (Larch *et al.* 2025a) as a different dataset. Second, we conduct several robustness checks concerning the treatment of missing values in the IMF DoTS dataset that was used to generate our baseline dataset. Third, we extend our main analysis to services trade using data from the International Trade and Production Database for Estimation (ITPD-E) (Borchert *et al.* 2021; Larch *et al.* 2025b).

6.1.1 | The WTO’s SGD

First, we use the WTO’s SGD (Larch *et al.* 2025a), which contains aggregate manufacturing trade covering the years 1980–2016.²⁷ The main differences relative to our main dataset are that the SGD includes domestic trade flows, covers a shorter time period, and only includes manufacturing trade rather than all merchandise trade as our baseline dataset.

The ETWFE results for the SGD are presented in Table 4, where the estimates are based on a small sample that contains 72 countries, accounting for 98% of world exports (columns (1)–(3)), a medium sample that contains 92 countries, accounting for 99% of world exports (columns (4)–(6)), and a large sample that contains all 229 countries (columns (7)–(9)). In addition, we capitalize on the fact that the SGD includes domestic trade flows, and we explore their importance for the intra-EU estimates.

We proceed in four steps. First, in column (1) of Table 4, we reproduce our main estimates, which are obtained with the small sample including domestic trade flows and with the full set of border-time dummies, which will account for common globalization trends. Second, in column (2), we use the same sample but without the common globalization trends. Consistent with the main argument from Bergstrand *et al.* (2015), the EU estimate is significantly larger. A possible explanation is that it now also captures some common globalization effects. Third, the results in column (3) are obtained from a sample without domestic trade flows. The estimate of the EU effects is positive and statistically significant, but smaller than the corresponding result from column (1). Consistent with Dai *et al.* (2014), the explanation for the difference is that the specification with domestic trade flows from column (1) enables us to explicitly account for diversion from domestic towards international sales. The intuition is that when two countries liberalize trade with each other, the producers in these countries find it optimal to sell more abroad and less domestically. Gravity estimations that are based only on international trade flows cannot capture such trade diversion effects.

We confirm these conclusions with our medium and large samples in columns (4)–(6) and (7)–(9) of Table 4, respectively. As our fourth step, we note that the coefficients in columns (1),

TABLE 4 Robustness with regard to a different dataset, domestic trade flows, and the construction of the baseline dataset.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
EU_{ijt}	0.328*** (0.033)	0.592*** (0.035)	0.076*** (0.036)	0.327*** (0.033)	0.592*** (0.035)	0.091*** (0.035)	0.326*** (0.033)	0.591*** (0.035)	0.102*** (0.034)	0.409*** (0.036)	0.376*** (0.036)	0.366*** (0.036)	0.234 (0.182)	0.255*** (0.048)
Dataset	SGD	SGD	SGD	SGD	SGD	SGD	SGD	SGD	SGD	Baseline	Baseline	Baseline	ITPD-E	ITPD-E
Sample	Small	Small	Small	Medium	Medium	Medium	Large	Large	Large	Small	Small	Small	Small	Small
Variant										V1	V2	V3	Services	Services
Estimator	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE	ETWFE
Observations	152,642	152,642	150,435	239,955	239,955	237,343	914,854	914,854	911,209	354,768	313,902	295,982	61,114	61,114
Exporters	72	72	72	92	92	92	229	229	229	90	90	90	86	86
Importers	72	72	72	92	92	92	229	229	229	90	89	89	86	86
Years	37	37	37	37	37	37	37	37	37	70	60	60	20	20
Coefficients	90	90	90	90	90	90	90	90	90	260	195	195	42	1
Exporter × importer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exporter × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cross-border × year FE	Yes			Yes			Yes			Yes				
Domestic trade	Yes	Yes		Yes	Yes		Yes	Yes						

Notes: The table presents PPML regression results from the ETWFE model (equation (1)), for which the cohort-year-specific treatment effects were aggregated to obtain an aggregate treatment effect estimate. Columns (1)–(9) are based on the WTO SGD, columns (10)–(12) are based on variants of our baseline dataset, and columns (13)–(14) are based on ITPD-E services data (Borchert *et al.* 2021; Larch *et al.* 2025b). The dependent variable is exports, which vary over the exporter–importer–year dimension. The ‘Small’ sample of the SGD contains 72 countries, accounting for 98% of world exports (columns (1)–(3)). The ‘Medium’ sample of the SGD contains 92 countries, accounting for 99% of world exports (columns (4)–(6)). The ‘Large’ sample of the SGD contains the full set of countries from the SGD (columns (7)–(9)). Variant ‘V1’ of our baseline dataset omits observations, in which missing values were replaced with zeros. Variant ‘V2’ of our baseline dataset additionally omits observations, in which data on exports were used to replace missing import data. Variant ‘V3’ of our baseline dataset additionally omits observations, in which UN COMTRADE imports were used to replace missing imports in the IMF data. Cross-border × year FE indicates whether border-time dummies are included, which account for common globalization trends. ‘Domestic trade’ indicates whether domestic trade flows are included. ‘Coefficients’ reports the number of estimated coefficients apart from the fixed effects. Standard errors in parentheses are clustered by country pair.

***, **, * indicate significance at the 1%, 5%, 10% level, respectively.

Stata code: `jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfe) fevar(idt_ci_idt_cj brdr_time) // Columns (1), (4) and (7); brdr_time are cross-border × year FE; jwddid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfe) fevar(idt_ci_idt_cj) // All other columnsestat simple, predict(xb) // All columns`

(4) and (7) are substantially smaller than those based on our baseline dataset. This is likely related to the other two salient differences between the datasets. First, the SGD does not include the 1957 and 1973 cohorts, which were found to have the largest treatment effects (see Figure 6). Second, the effects of manufacturing trade tend to be slightly smaller than those of trade in agricultural goods (Fontagné and Yotov 2024), which are also included in our dataset.

6.1.2 | Robustness checks related to the baseline dataset

Next, we consider a series of robustness checks concerning the treatment of missing values in the IMF DoTS dataset that was used to generate our baseline dataset. In columns (10)–(12) of Table 4, we replicate our main specification using the small baseline sample while progressively dropping observations associated with different assumptions imposed during the construction of our dataset.

First, in column (10) of Table 4, we omit observations in which missing values were replaced with zeros (60,975 out of 417,131). The resulting coefficient is very similar to our baseline estimate, but slightly smaller (0.409).

Second, in column (11) of Table 4, we additionally omit observations in which data on exports were used to replace missing import data (40,891 observations). This results in a coefficient that is large and positive, but smaller than our baseline estimate (0.373). Apart from changes in the control group, this is likely due to the omission of the 1957 cohort (note that the number of estimated coefficients drops from 260 to 195), which was associated with large treatment effects.

Finally, in column (12) of Table 4, we additionally omit observations in which UN COM-TRADE imports were used to replace missing imports in the IMF data (17,959 observations); that is, this variant essentially corresponds to the original IMF data on imports only. The resulting coefficient is similar to the one in column (11), albeit slightly smaller. Overall, we conclude that while slightly reducing the magnitude of the aggregate effect, the assumptions underlying the construction of our dataset do not materially affect our main result.

6.1.3 | Services trade

We also explore whether our main findings extend to services trade. For this exercise, we use the ITPD-E services data, which rely on administrative records. Because ITPD-E covers a shorter window, we re-estimate the core specification with both ETWFE and standard TWFE over 2000–2019. The ETWFE coefficient is 0.234 (standard error 0.182, not significant), while the TWFE estimate is 0.255 (standard error 0.048, significant at the 1% level). The point estimates are close, and their difference is statistically indistinguishable, given the imprecision of the ETWFE result. Given that differences between the ETWFE and TWFE models may arise from heterogeneity across cohorts and event time, and given the comparatively short sample period, it is perhaps not surprising that the estimates are not statistically different from each other.

6.2 | Degree of heterogeneity and alternative estimators

With regard to the heterogeneity of the estimator, we consider variants that are less flexible and that require the estimation of fewer coefficients by imposing restrictions on the treatment effect heterogeneity. In addition, we consider an imputation estimator that allows for arbitrary heterogeneity at the country-pair-year level. We also implement a jackknife estimator that can address potential IPPs, which may arise in non-linear models with fixed effects. Finally, we provide results for OLS instead of PPML. Our findings are reported in Table 5.

TABLE 5 Robustness with regard to degree of heterogeneity of the ETWFE estimator and alternative estimators

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EU_{ijt}	0.353*** (0.039)	0.409*** (0.034)	0.348*** (0.053)	0.328*** (0.063)	0.419*** (0.072)	0.422*** (0.063)	0.620*** (0.040)	0.532*** (0.037)
OLS/PPML	PPML	PPML	PPML	PPML	PPML	PPML	OLS	OLS
Estimator	ETWFE	ETWFE	Imputation	Jackknife TWFE	Jackknife ETWFE	ETWFE	ETWFE	TWFE
Unit heterogeneity		Cohort	Pair	Cohort	Cohort	Cohort	Cohort	Cohort
Time heterogeneity	Year		Year	Year	Year	Year	Year	Year
Standard error clustering	Exp × Imp	Exp × Imp				Exp, Imp, Year	Exp × Imp	Exp × Imp
Observations	417,131	417,131	417,187	417,131	417,131	417,131	356,204	356,204
Exporters	90	90	90	90	90	90	90	90
Importers	90	90	90	90	90	90	90	90
Years	70	70	70	70	70	70	70	70
Coefficients	65	8		260	260	260	260	1
Exporter × importer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exporter × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table presents PPML regression results (except columns (7) and (8), which are based on OLS) using variants of the ETWFE model (equation (1)), for which the cohort-year-specific treatment effects were aggregated to obtain an aggregate treatment effect estimate. The dependent variable is exports, which vary over the exporter–importer–year dimension. Column (1) imposes complete homogeneity along the cohort dimension, and only allows for event-year-specific treatment effects. Column (2) imposes complete homogeneity along the time dimension, and only allows for cohort-specific treatment effects. Column (3) shows results using an imputation estimator with the same fixed effects structure as the baseline; its standard error is based on 1000 bootstrap replications. Columns (4) and (5) show results for a jackknife TWFE and ETWFE using 1000 draws described in Nagengast and Yotov (2025). The regression in column (6) is the same as in column (2) of Table 3, but standard errors are clustered by exporter, importer and year. Columns (7) and (8) show results of the TWFE and ETWFE model estimated using OLS. ‘Coefficients’ reports the number of estimated coefficients apart from the fixed effects. Standard errors in parentheses are clustered by country pair (except for column (6)).

***, **, * indicate significance at the 1%, 5%, 10% level, respectively.

Stata code: `jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) hettype(event) // Column (1) jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) hettype(cohort) // Column (2) jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) cluster(exporter importer year) // Column (6) jwdid lntrade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) fevar(idt_ci_idt_cj) // Column (7); lntrade is defined as gen lntrade=ln(trade) jwdid lntrade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) fevar(idt_ci_idt_cj) hettype(twfe) // Column (8) estat simple, predict(xb) // Columns (1)–(2) and (6)–(8)`

6.2.1 | Restrictions on heterogeneity

First, we impose restrictions on the ETWFE model by allowing the treatment effect to vary only at the event-year level (column (1) of Table 5) or at the cohort level (column (2)). Note that the model in column (1) is akin to a standard event-study specification. We find that the resulting aggregate coefficient in column (1) is only slightly larger than the baseline TWFE estimate, but still substantially smaller than the corresponding ETWFE estimate. By contrast, the model that allows for only cohort heterogeneity in column (2) yields a coefficient that is very close, but still slightly smaller than our baseline ETWFE estimate. We conclude that cohort heterogeneity is more important in our setting than treatment effect heterogeneity across time.

6.2.2 | Imputation estimator

Next, we consider an imputation estimator that provides noisy estimates of individual treatment effects (Borusyak *et al.* 2024). While Wooldridge (2023) shows that the imputation approach and the ETWFE regression are numerically equivalent for PPML, this equivalence breaks down under the more complex fixed effects structure of the gravity setting (Nagengast and Yotov 2025). The imputation estimator applied to our dataset is also positive and significant (column (3) of Table 5), but smaller than our baseline ETWFE estimate.²⁸ The difference between the ETWFE and the imputation estimate likely stems from the fact that the imputation estimator estimates the fixed effects only using the control group, while the ETWFE estimator uses information on both the control and treatment groups.²⁹

6.2.3 | Jackknife bias correction

Now we consider potential IPPs, which may arise in non-linear estimators in gravity settings (Jochmans 2017; Pfaffermayr 2021; Weidner and Zylkin 2021). Nagengast and Yotov (2025) show using a Monte Carlo simulation that the standard errors of the ETWFE regression can be downward biased in the gravity setting, which might be remedied using a split-panel jackknife bias correction. Therefore here we jackknife estimates of both TWFE (column (4) of Table 5) and ETWFE models (column (5)). The resulting point estimates are virtually unchanged in both cases. The standard errors, however, both increase considerably. A formal statistical test (Z-test) of the difference between the TWFE and ETWFE coefficients that takes the covariance between the estimates into account yields *p*-value 0.1039.

6.2.4 | Alternative standard error clustering

In our next experiment, we investigate the robustness of our results to alternative clusterings of the standard errors. The results for our baseline estimates in Table 3 were clustered by country pair. Alternatively, the results in column (6) of Table 5 are clustered by exporter, importer and year. While the standard error almost doubles, the significance of the coefficient estimate remains essentially unchanged. We conclude that while alternative standard error clustering might affect the magnitude of the standard errors, the changes are not large and do not effect the significance of our main results.

6.2.5 | Ordinary least squares

Finally, we provide results using the OLS estimator instead of the PPML estimator. For comparison, columns (7) and (8) of Table 5 report results from an ETWFE model and a TWFE model, respectively. We draw two conclusions from these additional results. First, in our setting, the OLS estimator yields larger coefficients irrespective of the underlying model under consideration. Second, the estimated effect of the ETWFE model is still larger than that of the TWFE model. Therefore our main conclusion that the TWFE estimate is downward biased remains unchanged when considering the OLS estimator instead of the PPML estimator.

6.3 | DiD- and ETWFE-specific assumptions

With regard to DiD- and ETWFE-specific assumptions, we consider the effect of methodological choices related to the definition of treatment onset, the potential effect of the presence of other treatments, the exclusion of long-term effects, a variant relaxing the parallel trends assumption, and alternative weighting schemes for obtaining the aggregate treatment effect coefficient. Our findings are reported in Table 6.

6.3.1 | Treatment onset

Throughout the analysis, we assumed that the treatment ‘onset’ occurred two years before EU accession to capture anticipation effects. Here, we consider two alternative strategies. First, we follow Wooldridge (2021, 2023), who suggests omitting periods with anticipation effects such that the pre-treatment period then consists of only observations without anticipation effects. The resulting coefficient in column (1) of Table 6 equals 0.460—that is, it is slightly larger than our baseline estimate from column (1) of Table 3. This is as expected given that the anticipation effects were smaller than the remaining treatment effects. Second, we consider using the year of EU accession as the treatment onset by shifting the treatment ‘onset’ back again by two years. The anticipation effects then become part of the pre-treatment period. The corresponding coefficient in column (2) of Table 6 is close to our baseline estimate, albeit slightly larger. Overall, we conclude that our main results are robust to alternative approaches to addressing anticipation effects.

6.3.2 | Exclusion of other treatments

One potential concern when considering treatment effects over a long time period is that other policy interventions may be implemented in the meantime. As a result, one may confound the effects of these new policy interventions with the original treatment under consideration. In our case, the most salient event is the introduction of the European Monetary Union (EMU). To rule out any contamination from the euro, we omit all post-treatment periods of trade between EMU members from the estimation sample, taking into account the staggered entry dates of individual countries into the monetary union.³⁰ Column (3) of Table 6 shows that the resulting coefficient (0.401) is only very slightly smaller than our baseline estimate, suggesting that EMU membership does not have a significant impact on our main result.

TABLE 6 Robustness with regard to different DiD- and ETWFE-specific assumptions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
EU_{ijt}	0.460*** (0.039)	0.425*** (0.036)	0.401*** (0.033)	0.323*** (0.034)	0.434*** (0.040)	0.399*** (0.042)	0.536*** (0.053)	0.840*** (0.085)	0.665*** (0.069)	0.828*** (0.076)
<i>Sample/treatment</i>										
Omit anticipation periods	Yes									
Shift treatment by 2 years		Yes								
Exclude EMU			Yes							
Exclude effects > 20 years				Yes						
<i>Covariate interactions</i>										
In Distance					Yes					
<i>RTA & WTO controls</i>						Yes				
<i>Weights</i>						Obs	Cohort	Year	Cohort × year	Trade
Observations	414,355	415,743	411,815	412,711	415,589	370,431	417,131	417,131	417,131	417,131
Exporters	90	90	90	90	90	90	90	90	90	90
Importers	90	90	90	90	90	90	90	90	90	90
Years	70	70	70	70	70	70	70	70	70	70
Coefficients	244	244	239	142	589	262	260	260	260	260
Exporter × importer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exporter × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table presents PPML regression results using variants of the ETWFE model (equation (1)), for which the cohort-year-specific treatment effects were aggregated to obtain an aggregate treatment effect estimate. The dependent variable is exports, which vary over the exporter–importer–year dimension. Column (1) omits the year immediately before EU accession to exclude periods with anticipation effects, following Wooldridge (2023). Similarly, column (2) shifts EU accession forwards by one year. Column (3) omits all post-treatment periods of trade between EMU members. Column (4) excludes treatment effects 20 years after EU accession. Column (5) includes interactions between In Distance and cohort-year effects thereby relaxing the parallel trend assumption (Callaway and Sant’Anna 2021; Wooldridge 2023). Column (6) includes RTA and WTO dummies as exogenous control variables. Columns (7)–(10) report results for alternative weighting schemes for the cohort-year effects of column (2) of Table 3 that differ from the approach in all other specifications that give every post-treatment observation (‘Obs’) the same weight. Instead, the robustness checks give every cohort (column (7)), every event-year (column (8)) and every cohort-year (column (9)) the same weight, or use trade weights (column (10)). ‘Coefficients’ reports the number of estimated coefficients apart from the fixed effects. Standard errors in parentheses are clustered by country pair.

***, **, * indicate significance at the 1%, 5%, 10% level, respectively.

Stata code: `jwdid trade if exclude!=1, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) // Column (1); exclude equal to one in year 1 and 2 jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) 2 method(ppmlhdfc) fevar(idt_ci_idt_cj) // Column (2); FT_EU 2=FT_EU 2+2 for treated jwdid trade if EMU==0, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) // Column (3); EMU is a dummy indicating EMU membership jwdid trade if FT_EU==0 | year-FT_EU>20, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) // Column (4) jwdid trade DIST, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) // Column (5) jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) exovar(RTA WTO) // Column (6) jwdid trade, ivar(id_ci_cj) tvar(year) gvar(FT_EU) method(ppmlhdfc) fevar(idt_ci_idt_cj) // Columns (7)–(10) estat simple, predict(xb) // Columns (1)–(6) estat simple [pweight=w_g], predict(xb) // Column (7); w_g are cohort weights estat simple [pweight=w_t], predict(xb) // Column (8); w_t are event-year weights estat simple [pweight=w_gt], predict(xb) // Column (9); w_gt are cohort-year weights estat simple [pweight=w_trade], predict(xb) // Column (10); w_trade are trade weights`

6.3.3 | Exclusion of long-term effects

Our baseline estimate contains post-treatment effects for early cohorts that span several decades (see Figure 7). While common in the literature, this requires the parallel trends assumption to hold over a very long time period. As noted in the previous subsection, this also requires there to be no other treatments or policy interventions occurring in the meantime. If one is not willing to make these assumptions, then a simple solution is to exclude observations far away from treatment onset. In column (4) of Table 6, we exclude treatment effects 20 years after EU accession. The corresponding regression still yields a large and positive coefficient (0.323). As one would expect, however, this is substantially smaller than our baseline estimate given that we now exclude many periods that are associated with the largest treatment effects.

6.3.4 | Time-constant covariates

In Subsection 5.1, we did not find evidence for violations of the identifying assumptions apart from an anticipation effect in the year before EU accession. In other settings, the parallel trends assumption may hold only when conditioning on covariates. Therefore we next consider an extension of the ETWFE model in which the cohort-time-specific treatment effects are allowed to vary by time-constant covariates (Wooldridge 2023). This experiment is similar in spirit to the approach by Callaway and Sant'Anna (2021) who consider settings when the parallel trends assumption holds only after conditioning on observed covariates by using outcome regression, inverse probability weighting, and doubly-robust estimands. As time-constant covariate, we consider the bilateral distance between countries in column (5) of Table 6, which was the most salient difference between treated and never-treated country pairs (see Table 2), and which is also the most robust and widely used proxy for bilateral trade costs in the gravity literature (Anderson and van Wincoop 2004). The resulting coefficient is virtually unchanged. Note that this specification comes at the cost of estimating a considerably larger number of coefficients. We conclude that our main result is robust to alternative and less-demanding identifying assumptions.

6.3.5 | Regional trade agreement and WTO controls

Another potential concern is the presence of other major trade policy changes during our sample period. Since separately identifying the dynamic effects of multiple, overlapping treatments requires imposing additional identifying assumptions (de Chaisemartin and D'Haultfœuille 2023; Goldsmith-Pinkham *et al.* 2024), we augment our ETWFE specification in column (6) of Table 6 by including dummy variables for regional trade agreement (RTA) membership and WTO membership. Crucially, the ETWFE estimate of EU integration remains effectively unchanged. We therefore conclude that our main findings on the EU effect are robust to controlling for these other major trade policy changes.

6.3.6 | Alternative weighting schemes

Finally, we consider alternative weighting schemes of the cohort-year effects when computing the aggregate treatment effect. For our baseline, we give every post-treatment observation the same weight. For robustness, we give every cohort (column (7) of Table 6), every event-year (column (8)) and every cohort-year cell (column (9)) the same weight. In addition, we also consider a variant that uses trade weights (column (10)). However, all four alternatives yield aggregate treatment effects that are considerably larger than our baseline estimate since they place larger weights on

cohort-year cells with larger treatment effects. Therefore we conclude that our results are robust to using alternative weighting schemes.

7 | CONCLUSION

The EU is the world's largest single market area, and it is considered the most successful international integration effort to date. Yet there are surprisingly few studies that have attempted to quantify the effects of EU membership on the international trade among its members. Moreover, recent developments in the econometrics literature suggest that the estimates from the relatively few existing papers that estimate the impact of the EU single market on trade with standard TWFE techniques may be biased due to 'forbidden comparisons' that (mis)use already-treated units in the control group.

Against this backdrop, we make two contributions to the existing literature. First, we combine established methods from the structural gravity literature with recent heterogeneity-robust DiD methods, and we deploy several data sources to re-evaluate the impact of membership in the EU single market on international trade. Our results reveal that the effects of the EU have been very strong on average, and that the single market has benefited all of its members. In addition, we document certain asymmetries in the effects of the EU in favour of the exports from 'old' to 'new' members rather than in the opposite direction. These findings point to possible areas for policy intervention. We also perform a battery of robustness experiments and sensitivity checks, which broadly confirm and reinforce our main conclusions.

On the methods front, we introduce a new, fast and flexible estimation command that combines leading estimation techniques from the gravity literature with recent methods from the heterogeneity-robust DiD literature, and fast computation approaches to handle high-dimensional fixed effects in linear and non-linear settings. While our current analysis focuses on the effects of EU membership on trade, we expect that the impact and use of our command will be much broader. Naturally, it can be used to study the impact of other determinants of trade flows (e.g. WTO membership, currency unions) with the gravity model of trade. In addition, it can be used to estimate gravity equations for other bilateral flows and relationships (e.g. migration, FDI, patents, international M&A). Finally, it can be also be used in more general (non-gravity) panel and cross-sectional settings.

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ENDNOTES

¹ The ten countries that joined the EU in 2004 were Cyprus, the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia and Slovenia. See Figure 2 for details.

² See https://european-union.europa.eu/priorities-and-actions/actions-topic/trade_en (accessed 18 October 2025).

³ Consistent with the recent econometrics DiD literature (e.g. de Chaisemartin and D'Haultfœuille 2020; Callaway and Sant'Anna 2021; Goodman-Bacon 2021; Sun and Abraham 2021; Wooldridge 2021; Borusyak *et al.* 2024), we will refer to these traditional methods as 'two-way fixed effects (TWFE) estimators'.

⁴ EU integration entails multiple stages such as the founding EEC Treaty and the deepening of the single market, but our baseline estimate subsumes these into a single EU treatment coefficient. In the absence of sufficient variation

of these multiple treatments within and across cohorts, it is mathematically impossible to disentangle their dynamic effects without imposing further parametric assumptions on cohort- and event-time treatment patterns.

- ⁵ To install the `jwddid` command from SSC, use: `ssc install jwddid, replace`. To access the most up-to-date version, use: `net install jwddid, replace` from (<https://raw.githubusercontent.com/friosavila/stpackages/main>). Motivated by the relevant contributions of Professor Jeffrey Wooldridge, we named our new command `jwddid`, which stands for 'Jeffrey Wooldridge DiD'. To the best of our knowledge, `jwddid` is currently the only methodology that can simultaneously account for multiple fixed effects, as well as modelling non-linear outcomes (e.g. a Poisson model in our case). We refer the reader to de Chaisemartin and D'Haultfœuille (2022) and Roth *et al.* (2023) for surveys that compare the different DiD methodologies and how they fare in terms of efficiency, bias and assumptions, etc.
- ⁶ The sample ends in 2019 because of data availability; it was not chosen to accommodate or avoid Brexit. Were Brexit treated as part of the sample, it would represent an 'exit' and thereby violate the absorbing treatment assumption, introducing the sorts of biases in standard TWFE estimators documented in recent work (e.g. Hull 2018; Sun and Abraham 2021; de Chaisemartin and D'Haultfœuille 2023; Goldsmith-Pinkham *et al.* 2024). Wooldridge (2021, 2023) shows how the ETWFE framework can be extended to accommodate treatments with entry and exit by redefining cohorts based on both entry and exit years, and we refer to this approach and related work on sanctions with entry and exit (Kwon *et al.* 2024).
- ⁷ According to Eurostat (2023), 'Four fifths of total exports of goods within the EU in 2022 were manufactured products'.
- ⁸ We use the term 'structural' because gravity equation (1) is representative of many influential theoretical models of international trade (e.g. Eaton and Kortum 2002; Anderson and van Wincoop 2003; Arkolakis *et al.* 2012). However, we also note that the estimation techniques that we discuss and implement in this paper are much broader, and they apply to a wide range of gravity models for different types of bilateral trade flows, e.g. migration, FDI, patents, international M&A.
- ⁹ Consistent with theory, X_{ijt} may also include domestic trade flows. As summarized by Yotov (2022), using domestic trade flows in gravity regressions has significant implications for estimating the effects of various country-specific as well as bilateral policies, e.g. the EU. The intuition is that one of the main channels through which the EU has promoted international trade is at the expense of domestic sales. For robustness, we consider estimation results for the WTO SGD in Subsection 6.1, and our results are consistent with the existing literature.
- ¹⁰ Intuitively, the multilateral resistance terms capture the fact that trade between two countries depends not only on their sizes and the bilateral trade costs between them but also on how remote (geographically and economically) these countries are from the rest of the world. Yotov *et al.* (2016) summarize several attractive features of the structural multilateral resistance terms.
- ¹¹ Gravity model with exporter-time and importer-time fixed effects are often referred to as two-way gravity models. Similarly, those with exporter-time, importer-time and exporter–importer fixed effects are usually referred to as three-way gravity models. To avoid potential confusion with 'two-way fixed effects' models (i.e. with unit and time fixed effects) from the DiD literature, we refrain from using these terms in this paper.
- ¹² Some of the most recent gravity datasets, e.g. the International Trade and Production Database for Estimation of the US International Trade Commission, include more than 200 countries.
- ¹³ See, for example, de Chaisemartin and D'Haultfœuille (2023) for an approach to estimating heterogeneous treatment effects in the presence of multiple treatments in the linear setting. Similarly, see de Chaisemartin *et al.* (2022) and Callaway *et al.* (2024a,b) for estimation methods for continuous treatments in the linear setting.
- ¹⁴ To install the `jwddid` command from SSC, use: `ssc install jwddid, replace`. To access the most up-to-date version, use: `net install jwddid, replace` from (<https://raw.githubusercontent.com/friosavila/stpackages/main>).
- ¹⁵ Often, one can use group fixed effects instead of individual fixed effects, and would still obtain numerically identical results in the linear model case.
- ¹⁶ While the command does not impose the assumption that the data are a panel, the methodology is designed to work with panel data. In the case of repeated cross-sectional data, one should exclude `ivar(i)`.
- ¹⁷ The `jwddid` command has not been tested with all possible models. The user should be aware that the `method()` option is passed directly to the model estimation step, and the user should be familiar with the syntax of the model being estimated. Other 'methods' options can be used as well, following the syntax `method(cmd, cmdoptions)`.
- ¹⁸ Detailed information about the DoTS database can be found at <https://data.imf.org/?sk=9D6028D4-F14A-464C-A2F2-59B2CD424B85>, and for the COMTRADE database at <https://comtradeplus.un.org> (both accessed 19 October 2025).
- ¹⁹ The SGD is available at https://www.wto.org/english/res_e/reser_e/structural_gravity_e.htm (accessed 19 October 2025).
- ²⁰ The DGD can be downloaded at <https://catalog.data.gov/dataset/dynamic-gravity-dataset-1948-2016> (accessed 19 October 2025).
- ²¹ We include the UK in the analysis since it left the EU only in the year 2020, while our sample already ends in 2019.
- ²² Alternatively, using only untreated observations for the pre-treatment test in the spirit of Borusyak *et al.* (2024) yields very similar results (see Figure A1 in the Online Appendix).
- ²³ Note, however, that a joint significance test yields p -value 0.090, reflecting the small negative placebo effects in the years -4 and -2 .

²⁴ Calculated as $(e^{0.422} - 1) \times 100 = 52.5$.

²⁵ Consistent with this hypothesis, the PPML estimates from the robustness analysis of Mayer *et al.* (2019) are smaller than their OLS estimates. Similarly, the OLS results, which we obtain in our robustness analysis in Section 6, are significantly larger than the corresponding main PPML estimates.

²⁶ Note that this is not to say that Greece's trade has not benefited from its EU membership. In unreported country-specific results, we find that the EU effect for Greece's exports is indeed positive. This stems from Greece being part of subsequent cohorts (as an 'old' member).

²⁷ The SGD is available at https://www.wto.org/english/res_e/reser_e/structural_gravity_e.htm (accessed 19 October 2025).

²⁸ While we do not report standard errors for the imputation estimator, they could likely be obtained using a bootstrap procedure.

²⁹ See Nagengast and Yotov (2025) for a discussion.

³⁰ EMU entry dates are: 1999 for Austria, Belgium, Finland, France, Germany, Ireland, Italy, Luxembourg, the Netherlands, Portugal and Spain; 2001 for Greece; 2007 for Slovenia; 2008 for Cyprus and Malta; 2009 for Slovakia; 2011 for Estonia; 2014 for Latvia; 2015 for Lithuania.

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